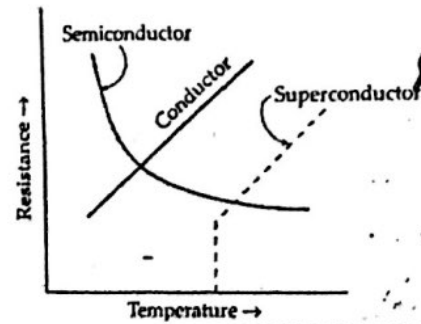


Chapter 5 (Unit-IV)

Part-1 Band Theory of Solids

- Introduction
- Kronig-Penney model, ✓
- E-k diagram,
- Effective mass of an electron, ✓



Part-2 Semi conductors

- Fermi level
- Intrinsic semiconductors, concentration of n in C.B & p in V.B,
- Extrinsic semiconductor: p-type & n-type semiconductors,
- Hall effect: Hall voltage and Hall coefficient. ✓

PPT slides by: Dr. Pragati sharma
Applied Physics-I

Introduction to theoretical developments

(1900) Classical free electron model of metals by Drude & Lorentz

According to this model, valence electrons are free to move throughout the volume of metal, forming an electron gas. "all free electrons contribute to the electrical conduction". It explains Electrical & Thermal conductivity of Metals. But cannot explain Electrical conductivity of Semiconductors & specific heat of metals.

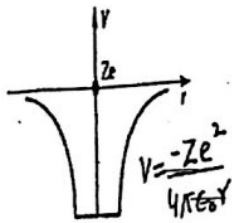
(1928) Quantum free electron model of metals by Sommerfeld

According to this model, valence electrons are free to move in zero potential field within the metal, but can not come out of the surface because an infinite potential exists at the surface (like particle in a box). "electrons near fermi surface can take part in electrical conductivity". It successfully explains properties of metals like specific heat, magnetic susceptibility, thermionic emission.

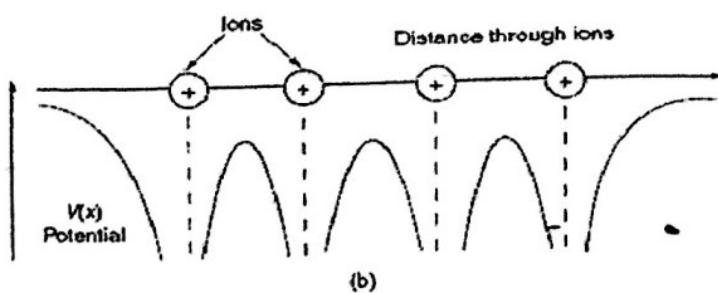
This theory fails to explain many other physical properties, e.g. (i) the difference between conductors, insulators and semiconductors, (ii) positive Hall coefficient of metals and (iii) lower conductivity of divalent metals than monovalent metals.

Modification in Quantum free electron model- Band theory of solids

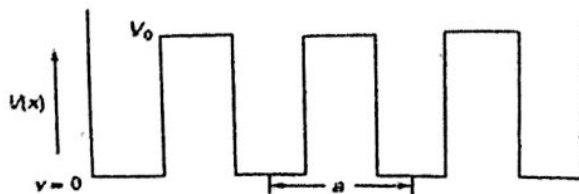
To overcome the above problems, electrons under the periodically varying potentials (due to the positive ions) in solids have been considered. If an electron moves through these ions, it experiences minimum potential at the ion site & maximum in between the two ion.



Band Theory of Solids could explain the allowed energies of electrons and the origin of energy gap in solids and hence could explain the difference b/n metals, insulators & semiconductors, and hence their electrical conduction.

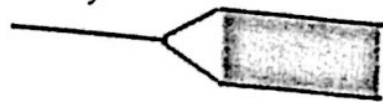


Actual Periodic Potential experienced by electron in 1D crystal lattice



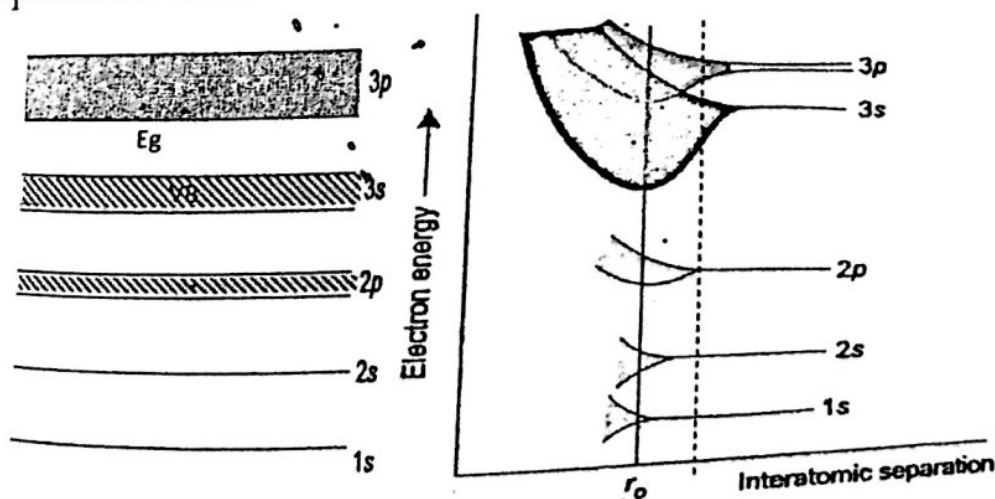
Idealized Periodic Potential by Kronig and Penney.

Qualitative Explanation of band theory (Appearance of bands in solids)



Isolated atoms do not interact with neighboring atoms, their electrons are tightly bound, they have discrete & sharp energy levels. N isolated atoms show n -fold degeneracy. When N isolated atoms are brought closer, their outermost orbits begin to overlap, electrons start interacting. At equilibrium distance, atoms are closely packed, intermixing of valence electrons among neighboring atoms leads to splitting of each interacting energy level into N energy levels. In solids, No. of atoms are very large $\sim 10^{23}$ atoms/mole, splitting of energy levels become so large & close together that they appears as continuous "band of energy levels".

The amount of splitting is different for different energy levels, as it depends on the extent of mutual influence of orbitals.

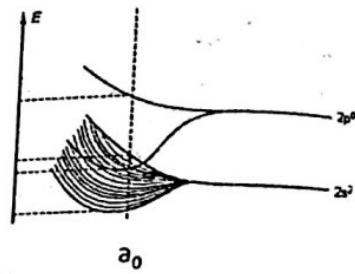
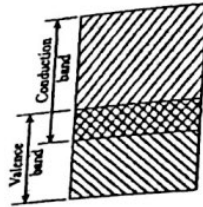


The highest occupied band is called Valence Band and empty band as Conduction Band. The regions separating highest occupied band from empty band is called Band gap or forbidden energy gap (E_g).

Conductor

(e.g. Be & Li)

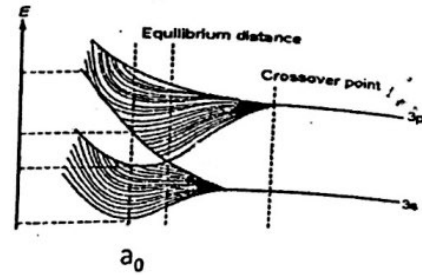
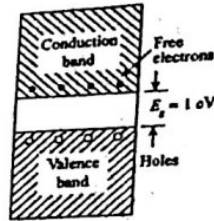
$$\rho < 10^{-3} \Omega \text{cm}$$



Semi-conductor

(e.g. Silicon)

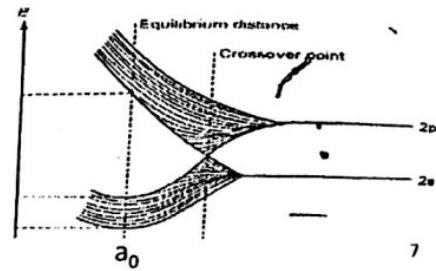
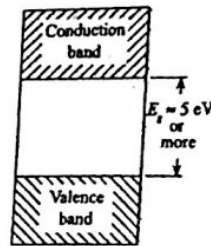
$$\rho = 10^{-3} - 10^6 \Omega \text{cm}$$



Insulator

(e.g. Diamond)

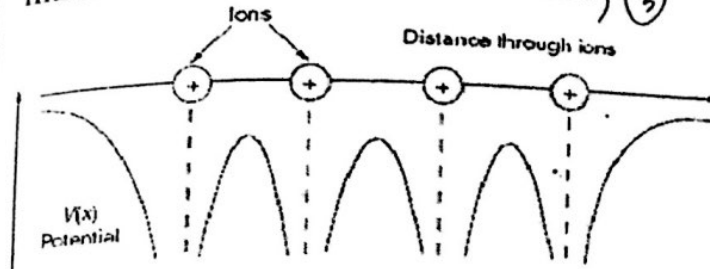
$$\rho > 10^6 \Omega \text{cm}$$



<u>Presence of energy bands, Gaps between them and Extent to which they are filled</u> governs the electrical behavior & other properties of solids. Accordingly solids are divided into three categories:			
Properties	Metals	Semi-conductors	Insulators
Energy gap	No gap or overlapping	$E_g \approx 1 \text{ eV}$	$E_g \geq 5 \text{ eV}$
Conductivity S/cm	$\sigma < 10^4 \text{ to } 10^7$	$\sigma = 10^{-6} \text{ to } 10^3$	$\sigma = 10^{-12} \text{ to } 10^{-18}$
Effect of temp on Resistivity	ρ increases linearly with temperature except at low temp.	ρ decreases exponentially with Temperature	ρ extremely high at low temperature

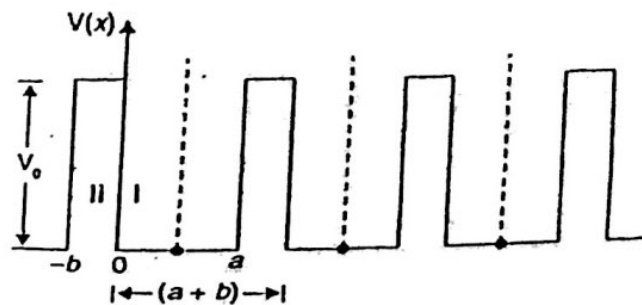
(Kronig-Penney model of band theory explains successfully the allowed energies of electrons & origin of energy gap in solids.)⁽¹⁾

(According to band theory, the electron in crystal lattice experiences a periodically varying potentials due to the positive ions in solids.) (It experiences minimum potential at the ion site & maximum in between the two ions.)⁽²⁾



Actual Periodic Potential experienced by electron in 1D crystal lattice

It is not easy to solve Schrödinger's equation with these potentials. So, Kronig-Penney considered the periodic potentials in the shape of rectangular steps i.e. square well-potential with period $(a+b)$.⁽³⁾



Idealized Periodic Potential by Kronig and Penney.

(Boundary conditions

✓ Well as Region I - $V(x)=0$ for $0 < x < a$ ⁽⁵⁾

✓ Barrier as Region II $V(x)=V_0$ for $-b < x < 0$ or $a < x < a+b$ ⁽⁶⁾

Where 'a' represents the width of square well & 'b' as width of potential barrier. The height of potential barrier is ' V_0 '. (This model is called Kronig-Penney model of potentials.)⁽⁷⁾

To find the allowed energies of electrons in crystalline solids, Kronig and Penney solved Schrodinger's equation in such a lattice.

$$\frac{\partial \psi}{\partial x} +$$

$$k^2 \psi$$

$$ik \frac{\partial \psi}{\partial x}$$

(The Schrödinger time-independent wave equation for the motion of electron along X-direction is given by

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{2m(E-V)}{\hbar^2} \psi = 0 \quad (8)$$

If 'E' represents Energy of electron in periodic potential $E < V_0$ and ψ_I & ψ_{II} as its wave functions in regions I & II respectively, then Time-Independent Schrodinger's equations for two regions:) (9

$$\left\{ \begin{array}{ll} \frac{\partial^2 \psi_I}{\partial x^2} + \frac{2mE}{\hbar^2} \psi_I = 0 & \text{for region I} \\ \frac{\partial^2 \psi_{II}}{\partial x^2} + \frac{2m(E-V_0)}{\hbar^2} \psi_{II} = 0 & \text{for region II} \end{array} \right. \quad (10)$$

$$\left(\text{Assuming } \alpha^2 = \frac{2mE}{\hbar^2} \text{ and } \beta^2 = \frac{2m(V_0 - E)}{\hbar^2} \right) \quad 11$$

$$\frac{\partial^2 \psi_I}{\partial x^2} + \alpha^2 \psi_I = 0 \quad \text{-----(1) for region I}$$

$$\frac{\partial^2 \psi_{II}}{\partial x^2} - \beta^2 \psi_{II} = 0 \quad \text{-----(2) for region II} \quad (11)$$

(Bloch theorem

In a crystal lattice, since potential varies periodically, the wave function of a particle should also vary periodically. According to F. Bloch, the Eigen function $\psi(x)$ of Schrodinger's wave equation in a periodic potential is given by

$$\psi(x) = e^{ikx} U_k(x) \quad \text{-----(3) known as Bloch function} \quad (13)$$

(It is a product of two functions, plane wave function e^{ikx} and periodic function $U_k(x)$ having a period same as that of a lattice

$$U_k(x) = U_k(x+a) = U_k(x+na) \quad (14)$$

(15) $\psi(x)$ represents a form of plane wave modulated with the periodicity of the lattice. Now differentiating eq. (3) twice w.r.t. x

$$\frac{\partial \psi}{\partial x} = ike^{ikx}U_k(x) + e^{ikx}\frac{\partial U_k}{\partial x} \quad (16)$$

$$\frac{\partial^2 \psi}{\partial x^2} = -k^2 e^{ikx}U_k(x) + 2ike^{ikx}\frac{\partial U_k}{\partial x} + e^{ikx}\frac{\partial^2 U_k}{\partial x^2} \quad \text{-----(4)}$$

Rewriting eq. (1) & (2) by applying Bloch theorem for region I & II (by substituting eq. 3 & 4)

$$\frac{\partial^2 U_I}{\partial x^2} + 2ik\frac{\partial U_I}{\partial x} + (\alpha^2 - k^2)U_I = 0 \quad \text{-----(5)} \quad (17)$$

$$\frac{\partial^2 U_{II}}{\partial x^2} + 2ik\frac{\partial U_{II}}{\partial x} - (\beta^2 + k^2)U_{II} = 0 \quad \text{-----(6)}$$

U_I & U_{II} represents the value of $U_k(x)$ in region I & II respectively. (18)

$$2ik\frac{\partial U_I}{\partial x}$$

$$2ik\frac{\partial U_{II}}{\partial x}$$

$$U_I = Ae^{i(\alpha-k)x} + Be^{-i(\alpha+k)x}$$

$$U_{II} = Ce^{(\beta-ik)x} + De^{-(\beta+ik)x}$$

Let the general solution of equation (5 & 6) are

$$U_I = Ae^{i(\alpha-k)x} + Be^{-i(\alpha+k)x} \quad \text{-----(7)}$$

$$U_{II} = Ce^{(\beta-ik)x} + De^{-(\beta+ik)x} \quad \text{-----(8)} \quad (19)$$

Value of constants A, B, C & D can be obtained by applying boundary conditions such as the requirement of continuity of wave function & its derivative between the two regions

$$\rightarrow [U_I]_{x=0} = [U_{II}]_{x=0} \quad \text{-----(9)}$$

$$\& \left[\frac{\partial U_I}{\partial x} \right]_{x=0} = \left[\frac{\partial U_{II}}{\partial x} \right]_{x=0} \quad \text{-----(10)}$$

$$\rightarrow [U_I]_{x=a} = [U_{II}]_{x=-b} \quad \text{-----(11)}$$

$$\& \left[\frac{\partial U_I}{\partial x} \right]_{x=a} = \left[\frac{\partial U_{II}}{\partial x} \right]_{x=-b} \quad \text{-----(12)} \quad (20)$$

$$[U_I]_{x=0} = [U_{II}]_{x=0}$$

This lead leads to four homogeneous equations.

$$A + B = C + D$$

$$i(\alpha - k)A - i(\alpha + k)B = (\beta - ik)C - (\beta + ik)D$$

$$e^{i(\alpha - k)a}A + e^{-i(\alpha + k)a}B = e^{-(\beta - ik)b}C + e^{(\beta + ik)b}D$$

$$i(\alpha - k)e^{i(\alpha - k)a}A - i(\alpha + k)e^{-i(\alpha + k)a}B$$

$$= (\beta - ik)e^{-(\beta - ik)b}C - (\beta + ik)e^{(\beta + ik)b}D$$

Determinant of coefficients of A, B, C & D equated to zero gives the solution of these equations.

$$\begin{vmatrix} 1 & 1 & 1 & 1 \\ i(\alpha - k) & i(\alpha + k) & (\beta - ik) & -(\beta + ik) \\ e^{i(\alpha - k)a} & e^{-i(\alpha + k)a} & e^{-(\beta - ik)b} & e^{(\beta + ik)b} \\ i(\alpha - k)e^{i(\alpha - k)a} & -i(\alpha + k)e^{-i(\alpha + k)a} & (\beta - ik)e^{-(\beta - ik)b} & (\beta + ik)e^{(\beta + ik)b} \end{vmatrix} = 0 \quad \text{---(13)}$$

$$\sinh(\alpha a) \sin(\beta b) + \cosh(\alpha a) \cos(\beta b) = \cos k(a + b)$$

The solution of Schrodinger wave equation comes out to be

$$\left(\frac{\beta^2 - \alpha^2}{2\alpha\beta} \right) \sinh(\beta b) \sin(\alpha a) + \cosh(\beta b) \cos(\alpha a) = \cos k(a + b) \quad \text{---(14)}$$

To obtain more convenient form of above equation, Kronig-Penney considered potential barrier to be a delta function, i.e. (22)

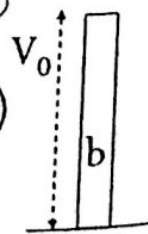
When $b \rightarrow 0$, then $V_0 \rightarrow \infty$, while (bV_0) remains finite.

The quantity $\lim_{\substack{V_0 \rightarrow \infty \\ b \rightarrow 0}} (bV_0)$ represents **barrier strength** (23)

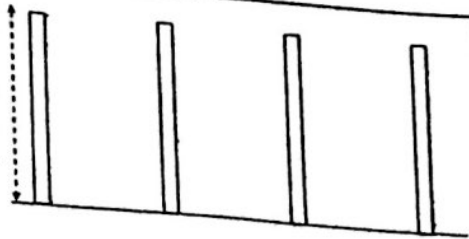
(For $b \rightarrow 0$;

$$\sinh(\beta b) \approx \beta b, \quad \cosh(\beta b) \approx 1$$

$$\cos k(a + b) = \cos ka \quad \text{---(24)}$$



(Under these assumptions, KP-model modifies to series of wells separated by infinitesimally thin potential barrier, eq. 14 simplifies to



$$\left(\frac{\beta^2 - \alpha^2}{2\alpha\beta}\right) \beta b \sin(\alpha a) + \cos(\alpha a) = \cos ka \quad \text{-----(15)}$$

Where $\beta^2 - \alpha^2 = \frac{2m(V_0 - E)}{\hbar^2} - \frac{2mE}{\hbar^2} = \frac{2m(V_0 - 2E)}{\hbar^2}$

$$\beta^2 - \alpha^2 \approx \frac{2mV_0}{\hbar^2} \quad \text{as } V_0 \gg E.$$

$$\left(\frac{\beta^2 - \alpha^2}{2\alpha}\right) b = \frac{2mV_0}{\hbar^2} \frac{1}{2\alpha}$$

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$$\left(\frac{mV_0}{\hbar^2} \frac{1}{\alpha}\right) \cdot b \sin(\alpha a) + \cos(\alpha a) = \cos ka \quad (26)$$

$$\left(\frac{mV_0 b a}{\hbar^2}\right) \cdot \frac{\sin(\alpha a)}{\alpha a} + \cos(\alpha a) = \cos ka \quad \text{-----(16)}$$

(Thus using Bloch theorem and all the boundary conditions for the continuity of the wave function the solution of Schrodinger wave eq for 1D periodic lattice comes out to be

$$\left[P \cdot \frac{\sin \alpha a}{\alpha a} + \cos \alpha a = \cos ka \right] \quad \text{-----(17)} \quad (27)$$

(Where $P = \left(\frac{mV_0 b a}{\hbar^2}\right)$ is referred as scattering power of a potential barrier. Its a measure of barrier strength ($V_0 b$). As P increases, barrier strength increases, electron is more tightly bound to a potential well (i.e. to the atom).) (28)

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In eq17

LHS has $P \cdot \frac{\sin \alpha a}{\alpha a} + \cos \alpha a$ which is a function of αa ,

(28)

where α is $= \sqrt{\frac{2mE}{\hbar^2}}$

RHS has $\cos ka$ which can have values ± 1 & -1 ...

(To satisfy eq17, only those values of $f(\alpha a)$ on LHS are acceptable which are lying between $+1 \leq \cos ka \leq -1$) (29)

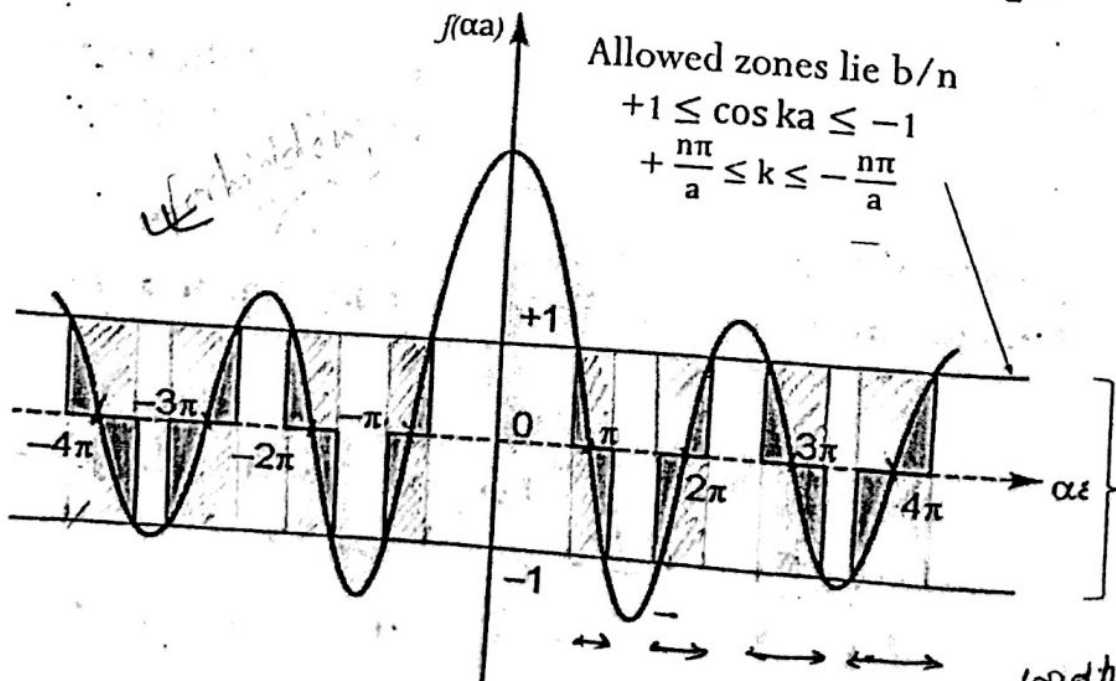
(Or only those values of α are acceptable (or energies are allowed) for which k values lie between or $+\frac{n\pi}{a} \leq k \leq -\frac{n\pi}{a}$) (30)

In other words energies of moving electrons for various values of k

$k = \pm \frac{n\pi}{a}$ for $n=1,2,3...$ are not permitted

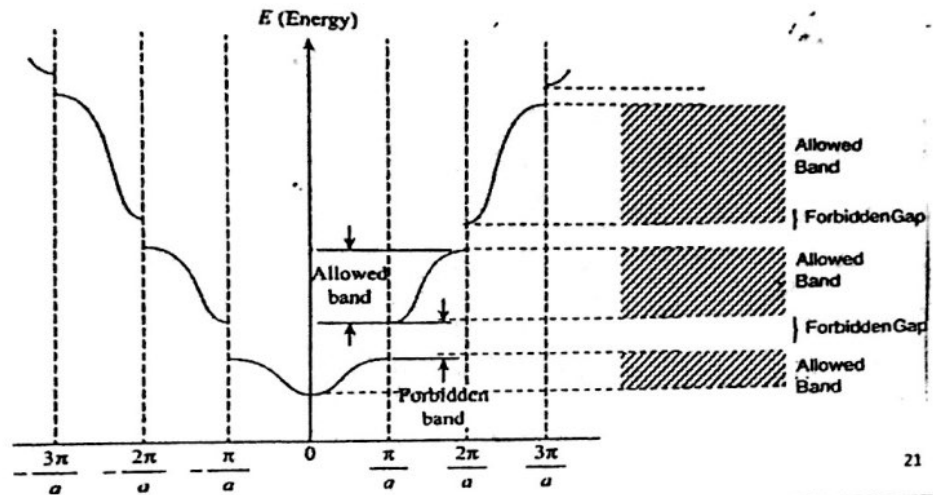
19

Variation of $\left(P \cdot \frac{\sin \alpha a}{\alpha a} + \cos \alpha a\right)$ with αa , for $P = \frac{3\pi}{2}$...



20

E-k curve: It demonstrates the allowed and forbidden energy bands for a particle travelling in a crystal lattice. The energy E versus wave number K for a particle in a crystal lattice can be plotted using $E = \frac{\hbar^2 k^2}{2m}$ and the boundary conditions of allowed bands, $+1 \leq \cos ka \leq -1$.

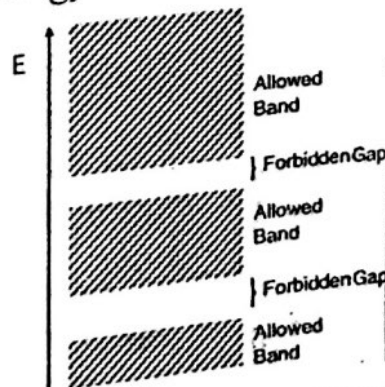


21

It shows interruptions in the parabolic curve at various values of k . The discontinuities in the energy in the $E-k$ curve appears at values of k lying at the boundaries given by $k = \pm \frac{n\pi}{a}$ for $n=1, 2, 3, \dots$. Thus

$$E = \frac{k^2 \hbar^2}{2m} \text{ at } k = \pm \frac{n\pi}{a} \text{ for } n=1, 2, 3, \dots \text{ are not permitted}$$

Hence the energy spectrum of electrons in a crystal is characterized by no. of allowed energy bands separated by forbidden gaps.



22

Conclusions from the graph in K-P Model:

$$P \cdot \frac{\sin \alpha a}{\alpha a} + \cos \alpha a = \cos ka \quad \text{---(17)}$$

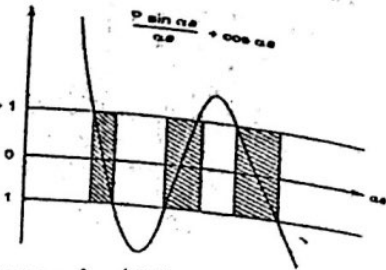
1. Energy spectrum of electrons in a periodic lattice is characterized by no. of allowed energy bands separated by forbidden zone.

2. Width of allowed band increases with increase in ' αa '

as ' αa ' increase, $[P \cdot \frac{\sin(\alpha a)}{\alpha a}]$ decreases,

Sharpness of curve reduces, becomes more flat,

causing the width of allowed band to increase



3. Width of allowed band decreases with increase in 'P'

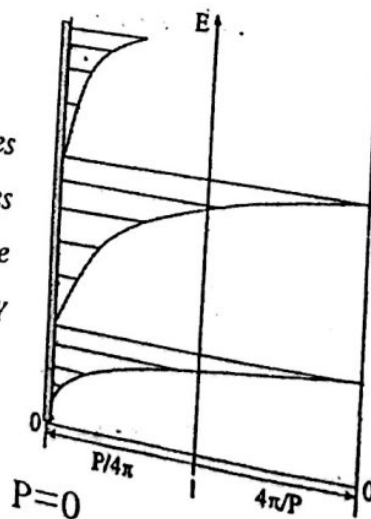
As P increases, $[P \cdot \frac{\sin(\alpha a)}{\alpha a}]$ crosses ± 1 at

steeper angle, width of allowed band decreases

23

Figure depicts the Physical significance of scattering power 'P' of a potential barrier. By varying P from 0 to ∞ , a completely free electron becomes completely bounded. Allowed energy bands are compressed into energy levels, as shown

At $P=0$,
Pot. barrier becomes
weak, electron moves
freely and will have
continuous
energy
spectrum.



At $P=\infty$,
electron is bounded
and will have line
spectrum i.e. discrete
Energy Levels.

$P=\infty$

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When $P \rightarrow 0$

Barrier Strength is weak

from eq. 17

$$\cos \alpha a = \cos ka$$

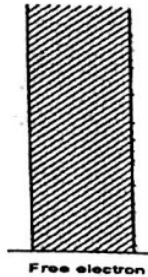
$$\alpha = k$$

$$\alpha^2 = k^2 \text{ gives}$$

$$\frac{2mE}{\hbar^2} = k^2$$

$$E = \frac{\hbar^2 k^2}{2m}$$

$$E = \frac{p^2}{2m}$$



It represents the equation of free electron. Energy spectrum is continuous.

When $P \rightarrow \infty$

Barrier Strength is large

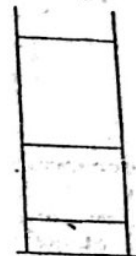
eq. 17 has a solution only if $\sin(\alpha a) \rightarrow 0$;

$$\alpha a = n\pi; n=1,2,3,\dots$$

$$\alpha^2 = \frac{n^2 \pi^2}{a^2} \text{ gives}$$

$$\frac{2mE}{\hbar^2} = \frac{n^2 \pi^2}{a^2}$$

$$E = \frac{n^2 \pi^2 \hbar^2}{2ma^2}; n=1,2,3,\dots$$



It represents the equation for an electron in an infinite potential well of width 'a'.

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Alternate explanation for the appearance of forbidden gaps in solids

The occurrence of gaps in the energy band diagram can be understood on the basis of Bragg's condition of diffraction

$$2a \sin \theta = n\lambda \quad n = \pm 1, \pm 2, \dots$$

where 'a' as spacing b/n the ions of the lattice & θ , angle of incidence

$$2a = n \frac{2\pi}{k} \text{ gives } k = \frac{n\pi}{a}$$

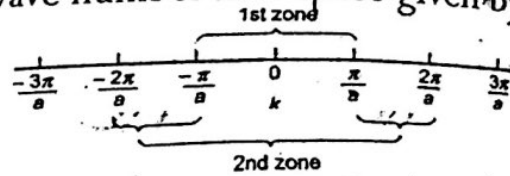
The gap at $k = \pm \frac{\pi}{a}$ arises because the wave reflected from one atom in the linear lattices interferes constructively with the wave reflected from the nearest neighbour atom. When the Bragg's condition is satisfied a wave travelling in one direction is soon Bragg-reflected & travel's in the opposite direction.

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✓ **Brillouin zone** represents the region in k space where electron has permissible values of wave vector k and hence the allowed values of energy E (in 1D, 2D or 3D crystal lattice).

Brillouin zone in 1D: According to Kronig and Penney, for a moving electron along 1-D periodic lattice, discontinuities in its energy appear at various values of wave number in k -space given by

$$k = \pm \frac{n\pi}{a}, \text{ where } n=1, 2, 3, \dots$$



First Brillouin zone region in k -space, between $(-\frac{\pi}{a} \& +\frac{\pi}{a})$, where electron has allowed energy

Second Brillouin zone lies between $(-\frac{2\pi}{a} \text{ to } -\frac{\pi}{a}) \& (+\frac{2\pi}{a} \text{ to } +\frac{\pi}{a})$

Similarly other higher Brillouin zones can be defined.

Brillouin zone in 2-D: In general, any discontinuity in 2-D lattice appears at

$$k_x n_1 + k_y n_2 = \frac{\pi}{a}(n_1^2 + n_2^2)$$

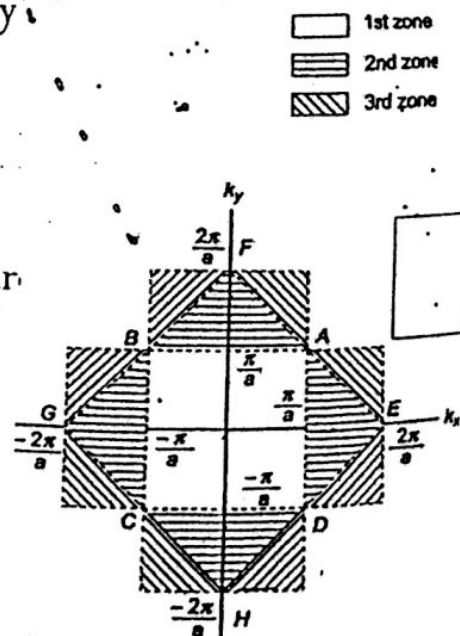
First Brillouin zone in XY plane is a square, the boundaries of which lies at

$$k_x = \pm \frac{\pi}{a} \text{ for } n_1 = \pm 1, n_2 = 0$$

$$k_y = \pm \frac{\pi}{a} \text{ for } n_1 = 0, n_2 = \pm 1$$

These four points encloses a square.

Second Brillouin zone is the regions shown & so on



Concept of effective mass: An electron in a crystal lattice is not free, it interacts with the lattice. Its behavior towards external force is different from that of a free electron. This deviation in the behavior of electron in crystal lattice can be taken into account by considering the electron to have an 'effective mass' m^* rather than mass m .

Effective Mass of electron

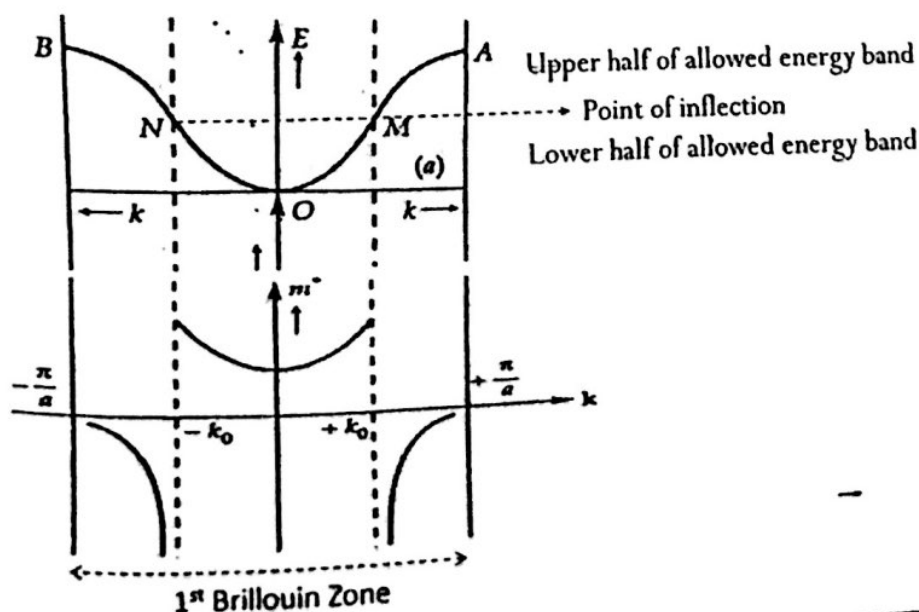
When an electron in a crystal lattice, under the influence of electric field or magnetic field, is accelerated, the mass of electron is found to be different from that of a free electron and is given by

$$m_e^* = \frac{\hbar^2}{d^2E/dk^2}$$

This called the effective mass of electron in a crystal lattice.

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The effective mass is not constant, it varies with the value of d^2E/dk^2 . In other words it depends on the location of electron in the allowed energy band.



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Derivation of Effective mass of electron

Consider an electron, which has a well defined mass m , under the influence of external field of strength E' for time dt . If velocity of moving electron is v_g , the distance travelled in time dt will be $v_g dt$. Work done in time dt is given by, $dE = \text{force} \times \text{displacement}$

$$dE = eE' v_g dt \quad \text{-----}(1)$$

Where v_g , according to de-Broglie is the average velocity with which particles propagate through a periodic potential

$$v_g = \frac{dE}{dp} = \frac{1}{h} \frac{dE}{dk} \quad \text{-----}(2)$$

Acceleration of this electron is given by

$$a = \frac{dv_g}{dt} = \frac{1}{h} \frac{d}{dt} \left(\frac{dE}{dk} \right)$$

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$$a = \frac{1}{h} \frac{d^2 E}{dk^2} \frac{dk}{dt} \quad \text{-----}(3)$$

The value for $\frac{dk}{dt}$ can be obtain by Substituting eq. (2) into eq. (1)

$$\frac{dk}{dt} = \frac{eE'}{h} \quad \text{-----}(4)$$

Substituting the eq. (4) into eq. (3) gives

$$a = \frac{1}{h} \frac{d^2 E}{dk^2} \frac{eE'}{h} = \frac{d^2 E}{dk^2} \frac{eE'}{h^2}$$

$$a = \frac{eE'}{h^2 \left(\frac{d^2 E}{dk^2} \right)^{-1}} \quad \text{-----}(5)$$

In Newtonian mechanics, the acceleration of electron under the influence of external field of strength E' is given by $a = \frac{eE'}{m}$. -----(6)

Comparing eq. (5) & eq. (6)

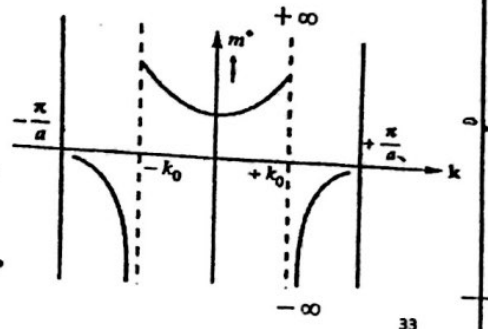
$$\frac{eE'}{m} = \frac{eE'}{\hbar^2 \left(\frac{d^2E}{dk^2} \right)^{-1}}$$

It is observed that the role of quantity $\hbar^2 \left(\frac{d^2E}{dk^2} \right)^{-1}$ in a periodic lattice is same as classical mass in Newtonian mechanics. This quantity is known as effective mass m^* , hence is given by

$$m^* = \frac{\hbar^2}{d^2E/dk^2}$$

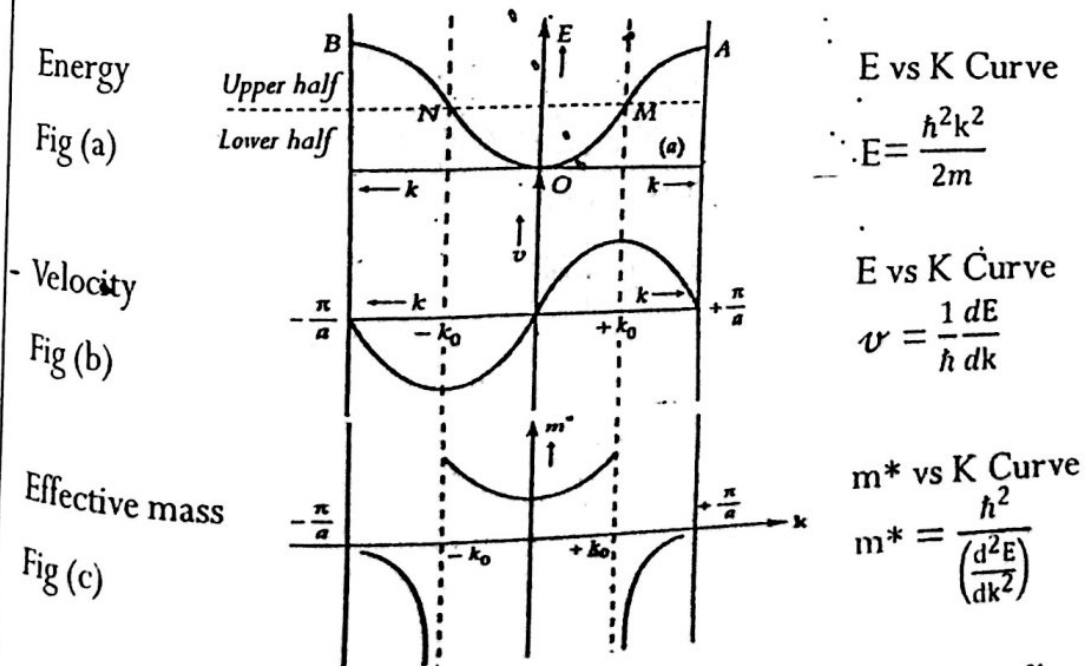
This shows that effective mass of electron in lattice is not constant

but it varies with the value of $\frac{d^2E}{dk^2}$



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Variation of effective mass as a function of k shows



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Fig (a) Variation of E with k -axis is symmetrical about E -axis having point of inflection at N & M . At O and at the boundary of Brillouin zone (A and B) the slope is $\frac{dE}{dk} = 0$. From point O to M or N , slope of E increases with increasing E . At point of inflection (M and N) slope of E , i.e. $\frac{dE}{dk}$ is maximum. Beyond this point, it decreases with increasing E .

Fig (b & c): At $k=0$, Lattice has no effect on the electron's response to Electric field (hence $m_e^* = m_e$). At the point of inflection, the velocity of electron reaches its maximum value, as slope dE/dk is maximum. The applied force causes no acceleration to the electron. (here m_e^* becomes infinity). The large effective mass can be interpreted due to strong binding force b/n the electron & the lattice.

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Fig (b & c).

Beyond the inflection point, the velocity of electron begins to decrease with increasing energy, particle starts decelerating i.e. acceleration is negative, ($d^2E/dk^2 < 0$). The effective mass m_e^* becomes negative. The concept of negative effective mass signifies that lattice exerts a retarding force on the electron, causing decrease in momentum. It can not pass freely into the crystal. This can be interpreted as Bragg's reflection when k is near $\pm\pi/a$. Reflection leads to gain in momentum in opposite direction. The Electron's response to the applied field in this k -region (k_0 to $\pm\pi/a$), is very different from that of a free electron. At $k = \pm\pi/a$, velocity of electron becomes zero, as it experiences a maximum retarding force.

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Concept of "hole"

The concept of "hole" can be explained beyond inflection point.

In the upper half of the energy band (k_0 to $\pm \pi/a$), the lattice exerts a retarding force on electrons, causing vacancies. In crystals which have energy bands nearly full except for a few vacancies near the tip of the band, these vacancies may be considered as +ve charge carriers called holes (-ve mass) to bring about conduction.

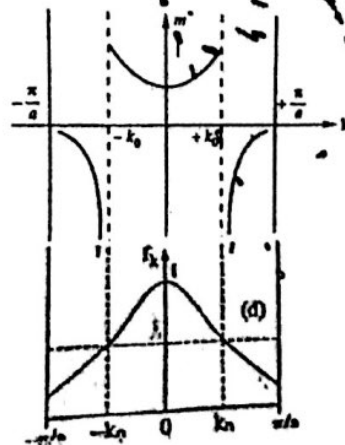
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Degree of freedom of an electron (f_k)

It measures the extent to which an electron is free in k -state & is defined as

$$f_k = \frac{m}{m^*}$$

$$f_k = \frac{m}{\hbar^2} \left(\frac{d^2 E}{dk^2} \right)$$

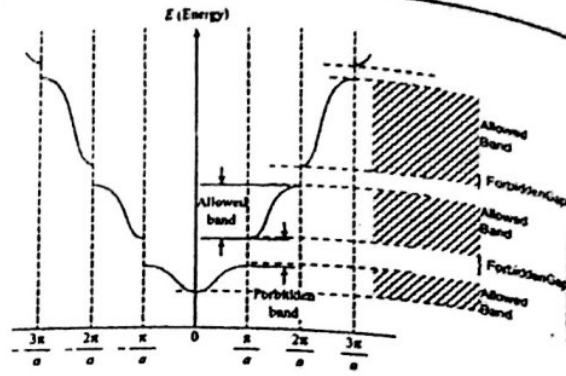
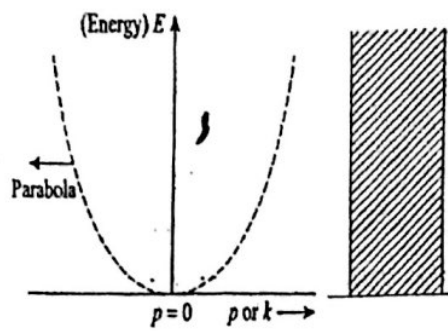


At $k=0$, $m^*=m$; & $f_k = 1$, electron behaves as free particle.

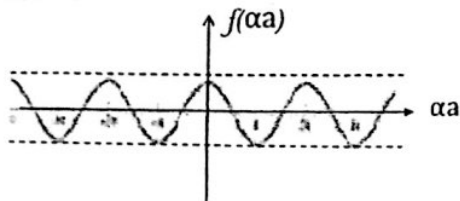
At $k=k_0$, m^* has a large value & so degree of freedom f_k is small.

The particle behaves as heavy particle, strong binding force exists between electrons & lattice.

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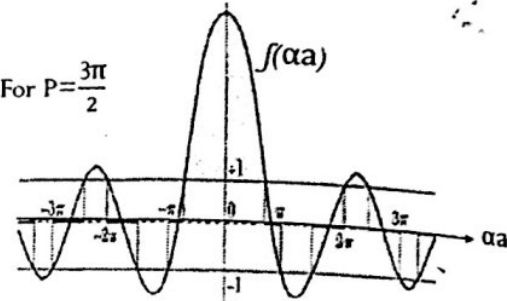


For $P=0$



- > Energies for all k values are allowed to electrons, i.e. electron is free.
- > Mass of free electron is m

For $P=\frac{3\pi}{2}$



- > Energies at $k = \pm \frac{n\pi}{a}$ for $n=1,2,3...$ are not permitted
- > Mass of electron is effective mass m^*

Q1. find the ratio between the kinetic energies of an electron in a 2-D Square lattice when (a) $k_x = k_y = \frac{\pi}{a}$ and (b) $k_x = \frac{\pi}{a}$ and $k_y = 0$

Solution: Knowing that inside the Brillouin zone, the energy of an electron is

$$E = \frac{\hbar^2 k^2}{2m} = \frac{\hbar^2}{8\pi^2 m} (k_x^2 + k_y^2) \quad \text{in 2-D}$$

$$(a) E_1 = \frac{\hbar^2}{8\pi^2 m} \left(\frac{2\pi^2}{a^2} \right) = \frac{\hbar^2}{4a^2 m}$$

$$(b) E_2 = \frac{\hbar^2}{8\pi^2 m} \left(\frac{\pi^2}{a^2} \right) = \frac{\hbar^2}{8a^2 m}$$

Hence the ratio is $\frac{E_1}{E_2} = 2$